11.31 Solution: Given E_ρ and B_z, $\mathbf{D}$ and $\mathbf{H}$ inside the cylinder can be expressed as

$$\mathbf{D}_{\text{in}} = \varepsilon\mathbf{E} + \gamma^2\left(\varepsilon - \frac{1}{\mu}\right)\left[\beta^2\mathbf{E}_\perp + \boldsymbol{\beta}\times\mathbf{B}\right] = \hat{\rho}\left[\gamma^2\left(\varepsilon - \frac{\beta^2}{\mu}\right)E_\rho + \beta\gamma^2\left(\varepsilon - \frac{1}{\mu}\right)B_z\right],$$

and

$$\mathbf{H}_{\text{in}} = \frac{\mathbf{B}}{\mu} + \gamma^2\left(\varepsilon - \frac{1}{\mu}\right)\left[-\beta^2\mathbf{B}_\perp + \boldsymbol{\beta}\times\mathbf{E}\right] = \hat{z}\left[-\gamma^2\left(\beta^2\varepsilon - \frac{1}{\mu}\right)B_z - \beta\gamma^2\left(\varepsilon - \frac{1}{\mu}\right)E_\rho\right].$$

Here, the velocity is in the $\hat{\phi}$ direction, $\mathbf{E} = E_\rho\hat{\rho}$ and $\mathbf{B} = B_z\hat{z}$ are perpendicular to it. Outside of the cylinder, $\mathbf{D}_{\text{out}} = 0$ and $\mathbf{H}_{\text{out}} = B_0\hat{z}$. From the continuity of the normal component of $\mathbf{D}$ and the tangent component of $\mathbf{B}$, we must have $D_\rho = 0$ and $H_z = B_0$. The first condition leads to

$$E_\rho = -\frac{\beta\left(\varepsilon - 1/\mu\right)B_z}{\varepsilon - \beta^2/\mu}.$$

Substitute this into the second condition, we will obtain

$$\left[-\left(\beta^2\varepsilon - \frac{1}{\mu}\right)B_z + \beta\left(\varepsilon - 1/\mu\right)\frac{\beta\left(\varepsilon - 1/\mu\right)B_z}{\varepsilon - \beta^2/\mu}\right]\gamma^2 B_z = B_0,$$

which gives B_z as

$$B_z = \mu B_0\frac{1-\beta^2/\mu\varepsilon}{1-\beta^2} = \mu B_0\frac{1-\omega^2\rho^2/(c^2\mu\varepsilon)}{1-\omega^2\rho^2/c^2}.$$

Then,

$$E_\rho = -\frac{\beta(1-1/(\mu\varepsilon))}{1-\beta^2/(\mu\varepsilon)}B_z = -\frac{\beta(1-1/(\mu\varepsilon))}{1-\beta^2}\mu B_0 = -\frac{\mu\omega\rho B_0}{c(1-\omega^2\rho^2/c^2)}\left(1 - \frac{1}{\mu\varepsilon}\right).$$

The voltage difference between the inner and outer surfaces is given by

$$\begin{aligned} V &= \left|\int_a^b E_\rho d\rho\right| = \frac{\mu\omega B_0}{c}\left(1 - \frac{1}{\mu\varepsilon}\right)\int_a^b \frac{\rho d\rho}{1-\omega^2\rho^2/c^2} \\ &= \frac{\mu\omega B_0}{c}\left(1 - \frac{1}{\mu\varepsilon}\right)\left(-\frac{c^2}{2\omega^2}\log\left(1-\omega^2\rho^2/c^2\right)\right)\Bigg|_a^b \\ &= \frac{\mu c B_0}{2\omega}\left(1 - \frac{1}{\mu\varepsilon}\right)\log\left(\frac{1-\omega^2a^2/c^2}{1-\omega^2b^2/c^2}\right). \end{aligned}$$

In the nonrelativistic limit, $\omega\rho \ll c$, we can approximate

$$\log\left(1-\omega^2\rho^2/c^2\right) \approx -\frac{\omega^2\rho^2}{c^2}.$$

Then, in this limit, the voltage difference becomes

$$V = \frac{\mu\omega B_0}{2c}\left(1 - \frac{1}{\mu\varepsilon}\right)(b^2 - a^2).$$